CS-3009 - Deliverable # 02 Faculty Feedback and Evaluation System

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Sprint 03 - Overview

1. **Duration:** 1 Week
2. **Sprint Goal:** To add department-based constraints and finalize the HOD, admin, teacher and student panel. Write core features for the HOD-panel, such as viewing instructors' performance and department growth.
3. **Modules Developed:**
    a. Database schema refinement
    b. Feedback form management (Create, Schedule)
    c. Teacher performance graphs.
    d. HOD dashboard.
    e. Student forms for feedback.

Sprint 03 - Backlog

a) User Stories

- As an admin, I want to view the history of which user accessed the system and for how much time.
- As a student, I want to respond on the live feedback form for reporting my concerns with the teacher and subject.
- As an admin, I want to add new questions to our question bank to be used in feedback forms.
- As an admin, I want to add and schedule new feedback forms.
- As an HOD, I want to view performance graph for overall department health and instructor-based growth report.
- As a teacher, I want to view my scores and track them across several semesters.

b) Sub-user Stories and Task Breakdown

User Story: Create Feedback Forms

- Design DB schema to save the forms.
- Create table for form responses and form scheduling.
- Implement UI for creating and scheduling feedback forms.
- List the created feedback form.
- Make it available for scheduling.

User Story: View Department Health.

- Write queries to get average scores for department across all performance factor.
- Choose suitable method to present the data visually.
- Write queries to get the average scores for department across all performance factor over several semesters.
- Ensure the score is calculated for current department only.

User Story: Create new questions

- Add different type of questions (Likert, Slider, Text, Binary)
- Store dynamic metadata for each type of question (Likert: Options, Text: Message for Student…)
- Link performance factor with the question for reports purposes.
- Test valid/invalid cases

User Story: HOD Panel

- Design basic UI layout
- Design Performance graphs.
- Dynamically update graphs based on updated feedback data.

Structured Specification:

User Story 1: Create Feedback Forms

Inputs	form_name, form_description, questions
Outputs	Confirmation of form creation
Processing	1. Validate Number of question >= 1 and Name or Description is not empty. 2. Insert the form information to the questionnaire table.
Preconditions	1. There must be at least 1 question in question bank. 2. All questions to be added in the questionnaire must exist in question bank.
Postconditions	Form is listed and available for scheduling.
Exceptions	1. Invalid input in form name or description 2. Try saving form with no questions.

User Story 2: Create New Questions

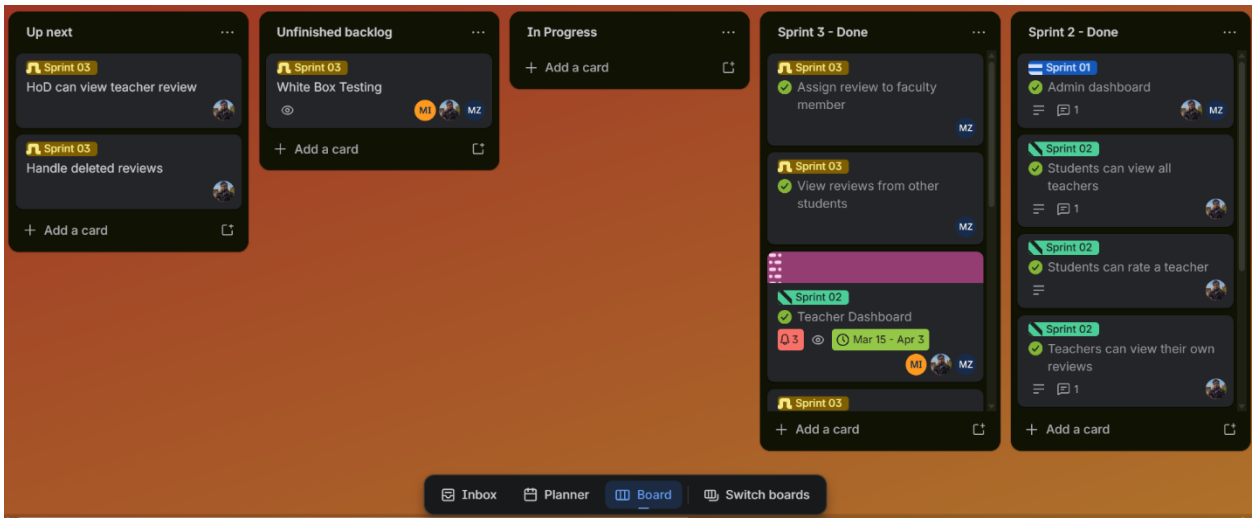
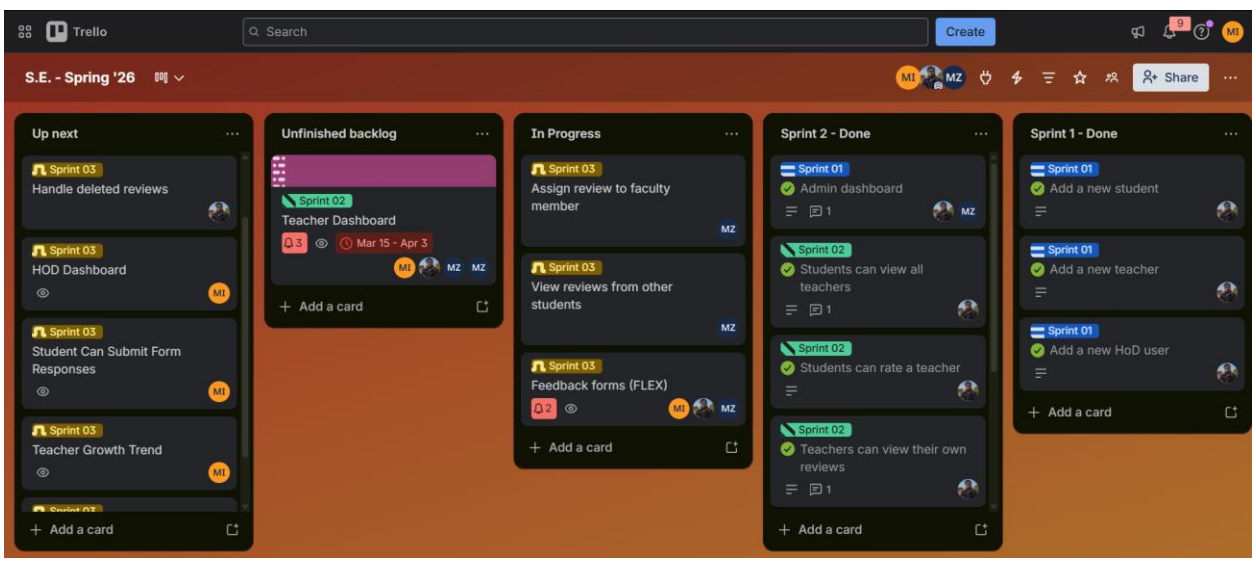
Inputs	Question Type, Question Factor, Question Metadata
Outputs	Confirmation of the question addition.
Processing	1. Validate the input fields for invalid values. 2. Insert Question to the question table.
Preconditions	1. Type of allowed questions must be configured. 2. Performance factors to be assigned to the questions must be configured.

Postconditions	Question is added to the question bank and is available to be used in creating forms.
Exceptions	1. Question text or metadata is empty. 2. Duplicate options added in Likert Type Questions 3. Max Slider value is less than Min Slider value in Slider Type Question. 4. Text input fields contain invalid inputs.

User Story 3: Schedule Feedback

Inputs	Form, department, course, semester, start date, end date
Outputs	Form is scheduled and will accept responses from start to end date.
Processing	1. Check if form, department, course existed previously. 2. Check if start and end date is within the semester duration, not before or after. 3. Insert scheduling information to the form schedules mapping table.
Preconditions	1. Form to be schedule must exist previously. 2. Department and Course for which to be scheduled must be present at time of form scheduling. 3. Semester for which to be scheduled is not from past.
Postconditions	Students can see and submit the form with appropriate repsones.
Exceptions	1. Scheduled date is out of semester bound. 2. Form, Department, or Course does not exist previously.

User Story 4: HOD access panel

Inputs	HOD login credentials
Outputs	HOD dashboard interface and all of its functionality
Processing	1. Authenticate HOD credentials 2. Fetch relevant data to be displayed in dashboard 3. Display the UI components (buttons, graphs, tabs…)
Preconditions	HOD must be registered and logged in
Postconditions	HOD have access to see and use HOD dashboard
Exceptions	1. Invalid Credentials 2. Data retrieval failure

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